TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	79.4 ± 2.33
Correlation Stability	81.8 ± 3.25
Distribution Consistency	88.8 ± 1.78
Entropy Preservation	90.9 ± 1.64
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	90.8 ± 3.34
Privacy Risk	50.5 ± 1.96
Data Density	100.0 ± 0.01
Coverage Score	20.0 ± 5.69
Feature Accuracy	86.9 ± 4.16
Novelty Score	0.4 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	claude-haiku-4-5
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	45.674	+0.928	2.1%	OK
TB	3.299	1.639	-1.659	50.3%	Drift
DB	1.486	0.675	-0.811	54.6%	Drift
Alkphos	290.576	275.878	-14.699	5.1%	OK
Sgpt	80.714	47.796	-32.917	40.8%	Drift
Sgot	109.911	61.410	-48.501	44.1%	Drift
TP	6.483	6.625	+0.142	2.2%	OK
ALB	3.142	3.309	+0.168	5.3%	OK
A/G Ratio	0.947	1.009	+0.062	6.5%	OK
Selector	1.286	1.448	+0.162	12.6%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	79.4 ± 2.33
Correlation Stability (*2)	81.8 ± 3.25
Distribution Consistency (*3)	88.8 ± 1.78
Entropy Preservation (*4)	90.9 ± 1.64
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	90.8 ± 3.34
Privacy Risk (*7)	50.5 ± 1.96
Data Density (*8)	100.0 ± 0.01
Coverage Score (*9)	20.0 ± 5.69

Feature Accuracy (*10)	86.9 ± 4.16
Novelty Score (*11)	0.4 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.670940170940171	0.6042735042735042	0.04529914529914532	*
F1	0.5587432119380917	0.729664735852335	0.6283649697462326	0.0696217578081409	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6333333333333332	0.6512820512820512	-0.03504273504273503	*
F1	0.7971299319768493	0.6973731340270064	0.7447836692238873	-0.05234626275296202	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.664957264957265	0.7170940170940171	0.01111111111111117	
F1	0.8102186993069884	0.7284863155760355	0.8052993649285302	-0.00491933437845828 5	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6512820512820513	0.6735042735042736	-0.01282051282051284 1	
F1	0.7989463929083696	0.7263212508246831	0.74992459638341	-0.04902179652495958	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.652991452991453	0.6888888888888889	-0.0170940170940171	
F1	0.8128002697439015	0.7225869167635411	0.7740757563540862	-0.03872451338981516	*

Model: SVM

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6213675213675213	0.6769230769230770	-0.02991452991452992	
F1	0.8279091624529537	0.6711961820531224	0.7550803822558053	-0.07282878019714843	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6555555555555557	0.6059829059829060	0.6675213675213676	0.01196581196581956	
F1	0.7596475050479288	0.6877829814034067	0.7616403400567829	0.00199283500885413 24	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6615384615384615	0.6888888888888889	0.01452991452991453	
F1	0.7821343746683586	0.729614095539078	0.7771492108464259	-0.00498516382193279 2	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6478632478632479	0.6991452991452991	-0.00341880341880340 67	
F1	0.7997903571549948	0.7179620981470413	0.7882030334145634	-0.01158732374043131	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6811965811965812	0.6487179487179486	0.6965811965811965	0.01538461538461539 6	
F1	0.7881789904780142	0.7152372451157507	0.7875877298983369	-0.00059126057967736 62	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	2.2	1.0	271	45	52	6.1	2.9	0.9
2	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0
3	54	Male	2.2	1.2	195	55	95	6.0	3.7	1.6
4	37	Female	0.8	0.2	205	31	36	9.2	4.6	1.0
5	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
6	18	Male	1.8	0.7	178	35	36	6.8	3.6	1.1
7	44	Male	0.9	0.3	218	38	42	6.5	3.2	1.0
8	55	Female	1.3	0.5	276	29	35	7.1	3.8	1.1
9	31	Male	0.7	0.2	194	47	51	6.2	3.1	0.95
10	68	Male	2.1	0.9	312	52	67	5.8	2.6	0.8
11	27	Female	0.6	0.1	157	19	22	7.6	4.2	1.2
12	52	Male	1.8	0.7	289	41	58	6.7	3.5	1.1
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	18	Male	0.6	0.1	265	97	161	5.9	3.1	1.1
2	34	Male	6.2	3.0	240	1680	850	7.2	4.0	1.2
3	22	Male	0.9	0.3	179	18	21	6.7	3.7	1.2
4	75	Male	2.5	1.2	375	85	68	6.4	2.9	0.8
5	29	Male	0.8	0.2	156	12	15	6.8	3.7	1.1
6	43	Male	1.3	0.6	155	15	20	8.0	4.0	1.0
7	41	Male	0.9	0.3	187	26	44	7.1	3.8	1.2
8	56	Female	1.5	0.7	412	67	89	6.3	2.9	0.8
9	28	Male	0.7	0.2	221	31	28	6.9	3.6	1.1
10	71	Male	3.2	1.4	356	71	95	5.8	2.5	0.7
11	44	Female	0.8	0.2	248	19	37	7.3	4.1	1.3
12	39	Male	1.1	0.4	267	42	61	6.4	3.2	0.9
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	5.7	2.8	214	412	850	7.3	3.2	0.78
2	38	Female	0.8	0.2	185	25	21	7.0	3.0	0.7
3	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
4	51	Male	0.8	0.2	367	42	18	5.2	2.0	0.6
5	58	Female	2.8	1.3	670	48	79	4.7	1.6	0.5
6	23	Female	2.3	0.8	509	28	44	6.9	2.9	0.7
7	52	Male	1.1	0.3	276	38	45	6.8	3.5	1.05
8	41	Female	0.9	0.2	198	28	31	7.2	3.8	1.15
9	67	Male	2.1	0.9	412	71	92	5.9	2.6	0.8
10	28	Female	0.6	0.1	147	19	23	7.4	4.1	1.3
11	59	Male	3.4	1.7	523	105	156	6.1	2.9	0.7
12	35	Male	1.5	0.5	284	44	52	6.9	3.6	1.0
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	34	Female	0.6	0.1	161	15	19	6.6	3.4	1.0
2	40	Male	1.2	0.6	204	23	27	7.6	4.0	1.1
3	35	Male	0.8	0.2	198	36	32	7.0	4.0	1.3
4	22	Male	0.8	0.2	198	20	26	6.8	3.9	1.3
5	38	Male	0.9	0.3	310	15	25	5.5	2.7	1.0
6	45	Female	1.0	0.3	250	48	44	8.6	4.3	1.0
7	51	Male	1.4	0.5	224	42	58	6.3	3.1	0.95
8	38	Female	0.7	0.2	178	28	26	7.2	3.7	1.05
9	68	Male	2.6	1.2	301	51	67	5.8	2.8	0.82
10	44	Female	0.9	0.3	189	35	31	7.4	3.9	1.15
11	57	Male	1.1	0.4	267	38	44	6.4	3.2	0.88
12	33	Female	0.6	0.1	155	24	22	6.9	3.6	1.08
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	34	Male	8.7	4.0	298	58	138	5.8	2.4	0.7
2	60	Male	2.0	1.1	664	52	104	6.0	2.1	0.53
3	58	Female	2.4	1.1	915	60	142	4.7	1.8	0.6
4	60	Male	4.0	1.9	238	119	350	7.1	3.3	0.8
5	70	Male	3.1	1.6	198	40	28	5.6	2.0	0.5
6	38	Male	1.7	1.0	180	18	34	7.2	3.6	1.0
7	52	Male	1.3	0.6	225	35	48	6.8	3.2	0.95
8	67	Female	2.5	1.2	312	42	89	6.1	2.3	0.55
9	41	Male	0.9	0.3	267	26	37	7.3	3.7	1.15
10	59	Male	5.2	2.8	445	67	156	5.9	2.6	0.62
11	35	Female	1.8	0.9	198	31	52	6.7	3.1	0.85
12	73	Male	1.2	0.4	289	28	41	6.2	2.9	0.75
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	55	Male	0.9	0.2	190	25	28	5.9	2.7	0.8
2	18	Male	0.9	0.3	300	30	48	8.0	4.0	1.0
3	13	Male	1.5	0.5	575	29	24	7.9	3.9	0.9
4	48	Female	0.8	0.2	218	32	28	5.2	2.5	0.9
5	55	Male	75.0	3.6	332	40	66	6.2	2.5	0.6
6	39	Male	3.8	1.5	298	102	630	7.1	3.3	0.8
7	51	Male	1.2	0.4	205	41	38	6.8	3.2	0.95
8	44	Female	0.9	0.2	178	28	31	7.4	3.9	1.1
9	67	Male	2.1	0.7	289	52	68	6.3	2.8	0.85
10	35	Female	0.6	0.1	192	19	22	6.1	3.5	1.3
11	58	Male	4.2	1.9	312	76	142	7.2	3.6	0.92
12	41	Male	1.1	0.3	167	34	29	5.9	2.7	0.88
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92
2	65	Male	4.9	2.7	190	33	71	7.1	2.9	0.7
3	58	Male	1.0	0.5	158	37	43	7.2	3.6	1.0
4	66	Male	11.3	5.6	1110	1250	4929	7.0	2.4	0.5
5	45	Female	0.8	0.2	165	22	18	8.2	4.1	1.0
6	58	Female	2.8	1.3	670	48	79	4.7	1.6	0.5
7	44	Male	1.2	0.4	210	35	41	6.9	3.5	1.0
8	37	Female	0.9	0.3	175	28	32	7.2	3.8	1.1
9	68	Male	3.2	1.5	380	55	78	6.4	2.8	0.7
10	26	Female	0.6	0.2	192	22	19	6.8	3.7	1.2
11	59	Male	2.4	1.1	420	67	91	6.1	2.4	0.6
12	41	Male	0.7	0.2	168	31	28	7.0	3.6	1.1
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
2	58	Female	0.8	0.2	130	24	25	7.0	4.0	1.3
3	60	Male	0.7	0.2	171	31	26	7.0	3.5	1.0
4	45	Male	1.1	0.4	92	91	188	7.2	3.8	1.11
5	40	Male	3.9	1.7	350	950	1500	6.7	3.8	1.3
6	14	Male	0.9	0.3	310	21	16	8.1	4.2	1.0
7	48	Male	1.2	0.4	218	52	61	6.9	3.5	1.08
8	36	Female	0.8	0.2	192	28	21	7.1	3.8	1.15
9	67	Male	6.4	3.2	402	89	156	6.2	2.8	0.75
10	29	Male	0.7	0.1	224	35	44	7.3	4.1	1.22
11	55	Female	1.5	0.7	267	118	201	5.9	2.6	0.82
12	41	Male	0.9	0.3	156	31	25	6.7	3.6	1.19
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	51	Male	0.8	0.2	160	34	20	6.9	3.7	1.1
2	33	Male	2.0	1.4	2110	48	89	6.2	3.0	0.9
3	38	Male	0.7	0.2	110	22	18	6.4	2.5	0.64
4	48	Female	1.1	0.7	527	178	250	8.0	4.2	1.1
5	48	Male	5.0	2.6	555	284	190	6.5	3.3	1.0
6	45	Male	0.6	0.2	245	22	24	7.1	3.4	0.9
7	44	Male	1.1	0.3	215	29	31	6.5	3.2	0.95
8	52	Female	0.9	0.2	186	35	28	6.8	3.6	1.08
9	39	Male	2.3	1.1	441	67	118	6.2	2.9	0.85
10	68	Male	1.5	0.6	298	41	57	5.8	2.4	0.72
11	31	Female	0.7	0.1	145	28	24	7.4	4.0	1.2
12	56	Male	3.4	1.8	512	89	156	6.1	2.6	0.68
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	47	Male	0.9	0.2	265	40	28	8.0	4.0	1.0
2	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
3	61	Male	1.5	0.6	196	61	85	6.7	3.8	1.3
4	40	Male	3.9	1.7	350	950	1500	6.7	3.8	1.3
5	42	Female	7.4	3.6	298	52	102	4.6	1.9	0.7
6	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
7	52	Male	1.2	0.4	245	38	42	6.9	3.6	1.1
8	43	Female	0.8	0.2	198	28	25	7.4	4.1	1.3
9	67	Male	2.1	0.9	312	52	89	6.4	3.0	0.85
10	39	Male	1.4	0.5	276	67	124	6.8	3.5	0.98
11	54	Female	0.9	0.3	215	35	31	7.1	3.8	1.2
12	41	Male	3.2	1.6	428	156	287	6.2	2.6	0.72
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3